# MEGHA JOSHI

MSc Data Science Student | SQL, Python, Transaction Monitoring, Data Visualisation
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## PROFILE

MSc Data Science student at the University of Glasgow with a computer science degree. I built an anti-money-laundering transaction monitoring system: on 60,059 transactions it cut false alerts by 98% while still catching every planted laundering case, and its SQL and Python rules are checked against each other by a 67-test suite in CI. I build data pipelines, check data quality, use statistics to find what matters, and explain it through dashboards to people without a technical background. Looking for a graduate role in data analytics, data engineering or data science.

## EDUCATION

**Master of Science (MSc) in Data Science, University of Glasgow, UK** Sep 2025 - 2027 (expected)
Modules: Statistical Methods, Information Visualisation, Machine Learning, Big Data Systems (Hadoop, Spark/PySpark), Text and Information Retrieval, Deep Learning, Large Language Model Principles

**B.E. Computer Science and Engineering, Chandigarh University, India** Aug 2020 - Jun 2024
Coursework: Database Systems, Data Structures and Algorithms, Object-Oriented Programming, Software Engineering

## TECHNICAL SKILLS

**Financial Crime Analytics:** Anti-money laundering transaction monitoring, alert tuning and false positive reduction, anomaly detection, testing rules against changed launderer behaviour
**Data Analysis:** SQL (joins, subqueries, window functions), Python (pandas, NumPy, SciPy), data pipelines (ETL), data cleaning and data quality checks, blending data from multiple sources, MS Excel (pivot tables, charts)
**Statistics:** Hypothesis testing, regression, confidence intervals, trend and seasonal analysis
**Visualisation and Tools:** Plotly Dash and Streamlit dashboards, matplotlib, Tableau and Power BI (currently learning), PySpark, Git, pytest, GitHub Actions CI

## WORK EXPERIENCE

**Junior Embedded Engineer, Radius Synergies International Pvt. Ltd., Noida, India** Dec 2024 - Jun 2025
- Tested and validated utility meters and source code with a colleague before deployment, and wrote the technical documentation and test reports used in QA review.
- Wrote firmware in Embedded C for pre-metering systems on STM32G070, STM32F101CB, LPC812 and LM3S815 microcontrollers.

## PROJECTS

**AML Transaction Monitoring** | Python, SQL github.com/meghajoshi-ds/AML-Transaction-Monitoring
- On 60,059 synthetic transactions, combined four detection rules with an Isolation Forest. False positives fell from 5,245 to 89 with all 139 planted laundering cases still caught. Alert precision rose from 2.6% to 61%, which saves about 1,290 analyst hours a year.
- Tested what happens when a launderer changes behaviour to get around the rules: the rules alone went from 139 cases caught to 14, while the anomaly model still caught 44.
- Rewrote the rules in SQL with window functions, with tests that check SQL and Python flag exactly the same transactions. The 67-test suite runs in CI on pandas 2 and 3.

**NHS Waiting Times Data Pipeline and Dashboard** | Python, SQL, Plotly Dash github.com/meghajoshi-ds/nhs-waiting-times
- Built an ETL pipeline in Python that extracts over five years of NHS England A&E and referral-to-treatment data (237 trusts, 141,504 RTT rows) plus NHS Scotland data, then cleans, standardises and loads both sources into one SQLite database.
- Built an interactive dashboard for a non-technical audience: a patient picks a specialty and region, then sees the fastest providers and how many weeks they could save by switching.
- Used a Mann-Kendall trend test with Sen's slope and seasonal decomposition to show that 4-hour A&E performance has declined significantly ($p < 0.001$), and that trusts stronger on emergency care also do better on elective care ($r = 0.37$, 124 trusts).

**Credit Risk Scorecard** | Python, LightGBM, SHAP, Streamlit github.com/meghajoshi-ds/credit-risk-scorecard
- Built logistic regression and LightGBM models to predict default on 163,987 Lending Club loans (AUC 0.690, KS 0.283) and found that interest rate leaks the answer, as the lender sets it from its own risk model.
- Turned the model into a 7-characteristic WoE points scorecard and a Streamlit app that scores an applicant and explains the main reasons behind the score.

## ACHIEVEMENTS, CERTIFICATIONS AND PUBLICATIONS

- CodeOlympics 2026 Gold Medal: 1st place as part of Team 29 (1,884.6 points), Glasgow University Tech Society
- Kaggle Bronze Medal: Kaggriculture competition, ranked 1,007th of 10,246 teams (top 10%)
- Certifications: Microsoft Certified: Azure Fundamentals (AZ-900); Software Testing (NPTEL); Fundamentals of Java (Infosys Springboard)
- Publication (peer-reviewed): Tokenizing Art: Unpacking the Impact of Non-Fungible Tokens on the Ownership Paradigm, IEEE Xplore, 2024